

Student Marketplace

Five Questions Solution with Excalidraw Diagrams

This version includes one editable Excalidraw source diagram for each planning question. The PDF renders the diagrams visually, while the source files remain in the diagrams folder.

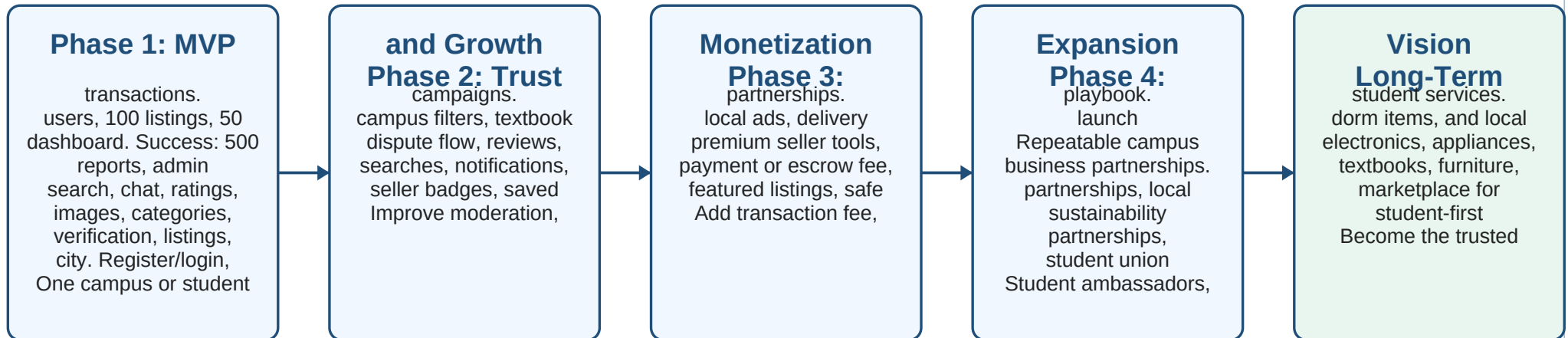
Core answer

Build a focused campus marketplace MVP that proves demand, trust, and repeat use before expanding. Start with textbook resale and student essentials, then add secure payments, delivery partnerships, and university/community integrations.

1. PHASES - What is the MVP? What comes after? What is the long-term vision?

The product should be built in phases. Start with one trusted student marketplace in one focused area, then grow trust, monetization, and campus expansion.

1. PHASES - MVP, Next Steps, Long-Term Vision



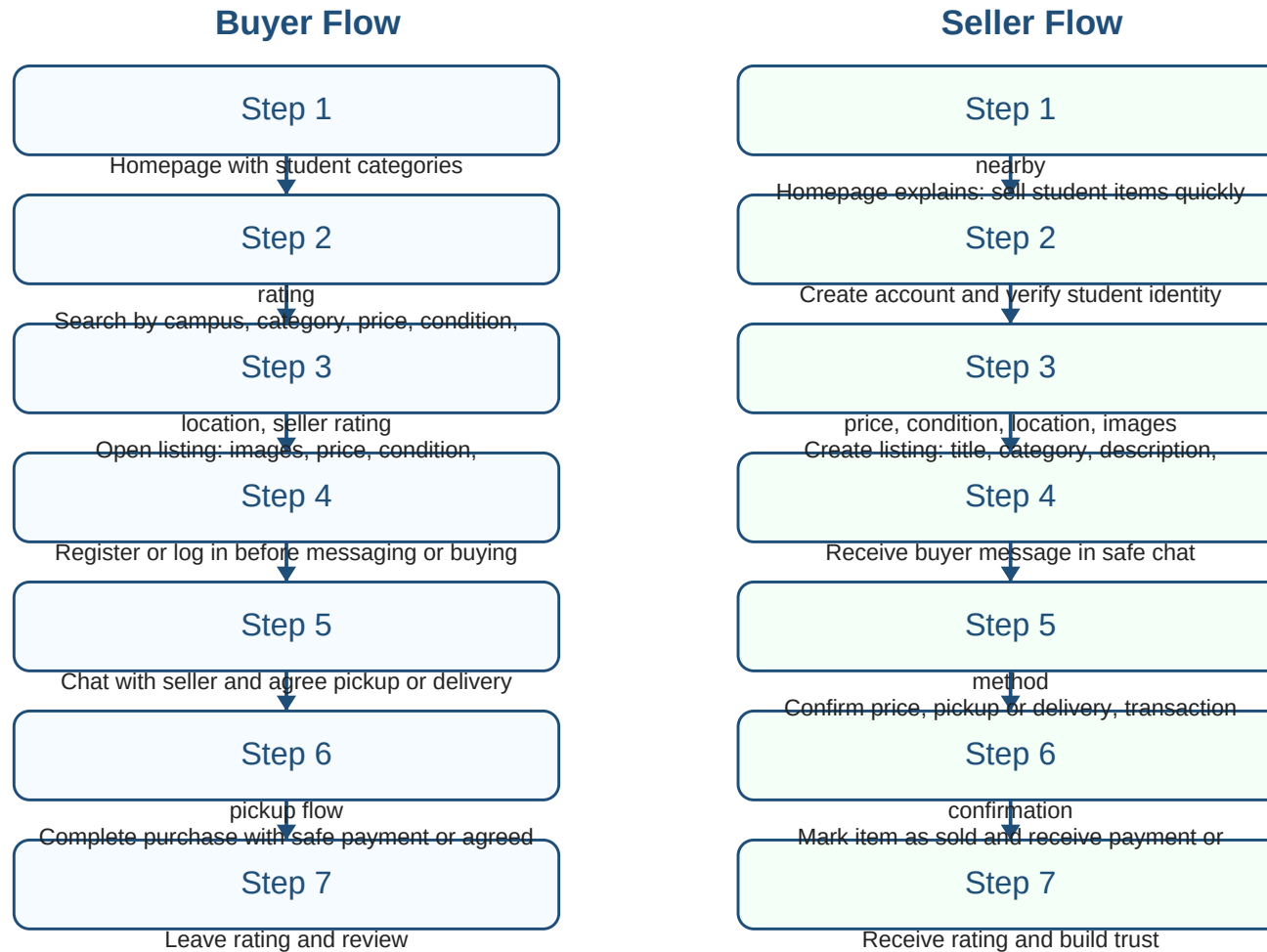
heavy city.

Decision: The MVP is not eBay for everyone. It is a trusted student marketplace for one campus or one student-

2. USER FLOW - What does the user actually do from landing to first success?

The first success should happen quickly: a seller posts a useful item, and a buyer finds it, contacts the seller, and completes a safe purchase.

2. USER FLOW - Landing to First Success

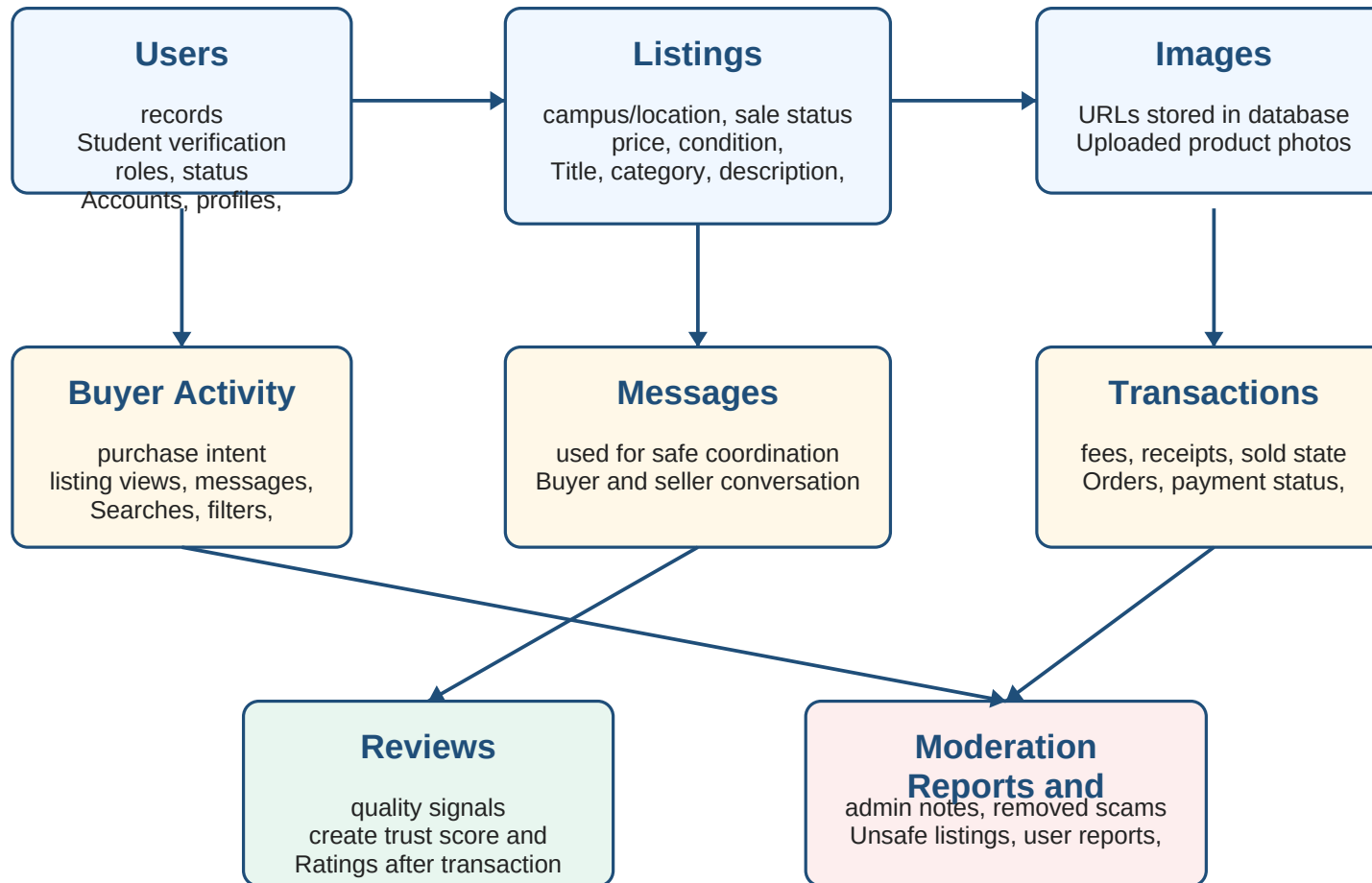


item safely inside the campus community.
First success: A student lists or buys one useful

3. DATA FLOW - What data is created, stored, moved, or transformed?

The marketplace creates and moves data around users, listings, images, search, messages, payments, reviews, and moderation. Trust and safety data is core product data.

3. DATA FLOW - Created, Stored, Moved, Transformed

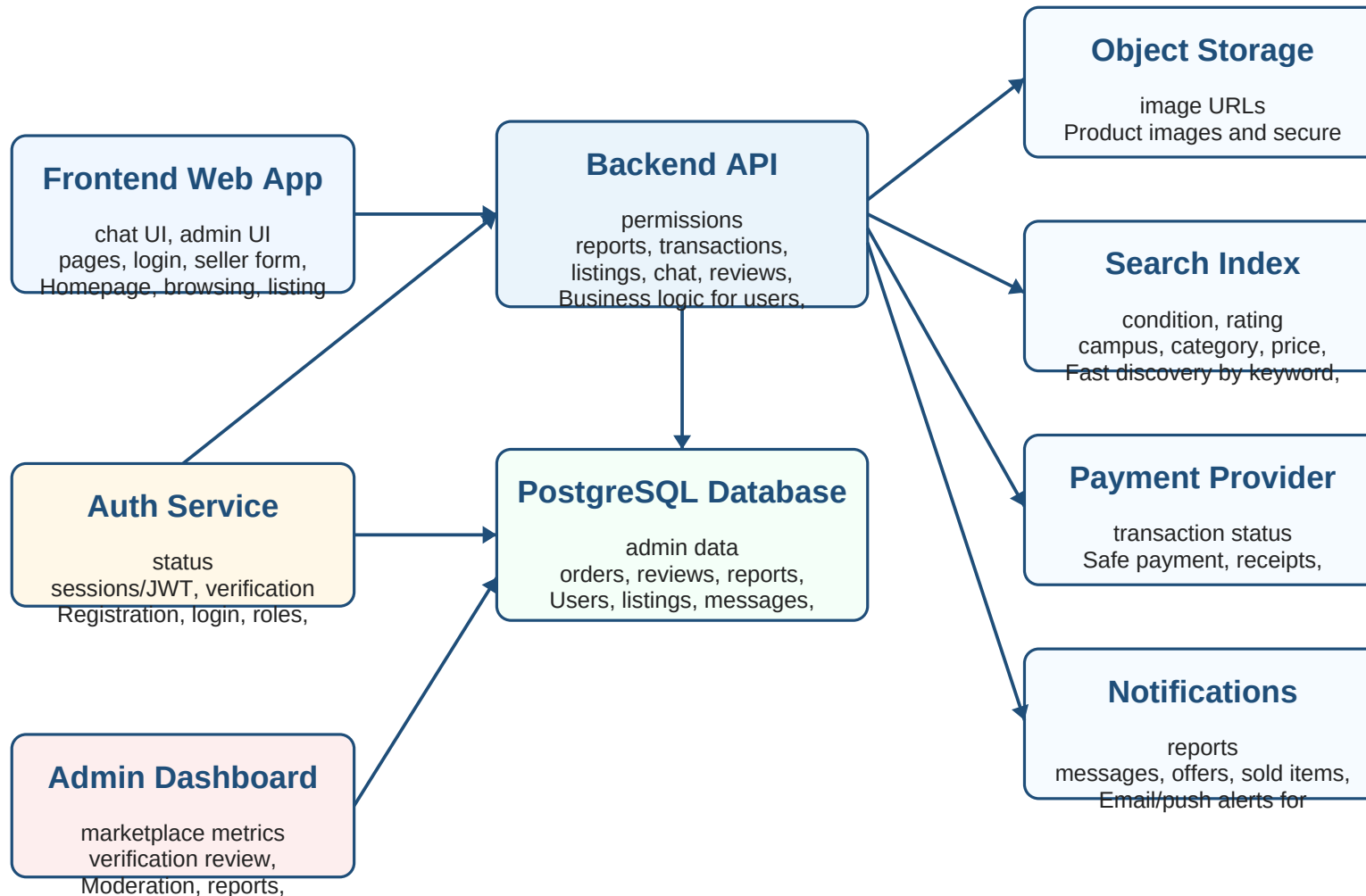


Trust and safety data is core product data, not an optional add-on.

4. ARCHITECTURE - What are the components? How do they connect?

The MVP architecture should support a web/mobile frontend, backend API, database, file storage, search, notifications, payments, and admin tools.

4. ARCHITECTURE - Components and Connections



search, payments, notifications, and admin tools connect around the API.

Flow: User opens frontend -> frontend calls backend API -> API reads/writes database -> images,

5. STACK RATIONALE - Why this framework, database, and AI model?

Choose technology because it supports the business goal: fast to build, secure enough for real users, simple to operate, and scalable later.

5. STACK RATIONALE - Framework, Database, AI Model

Layer	Recommended Choice	Why This Choice	Why Not First
Frontend	Next.js or React	strong ecosystem. components, SEO for listings, Fast marketplace UI, reusable	responsive web first. expensive for MVP. Start Native mobile is slower and more
Backend	Node.js/NestJS ASP.NET Core Web API or	team. Core is excellent for a C# transactions, scaling. ASP.NET Good for APIs, auth,	clean modular monolith. Avoid microservices early. Use a
Database	PostgreSQL	reports, transactions. listings, orders, reviews, Relational integrity for users,	relational structure. trust data benefits from NoSQL can work, but marketplace
Images	Cloudinary Azure Blob, S3, or	designed for product images. Cheap, scalable object storage	the main database. Do not store images directly in
Search	Meilisearch later first; Algolia or PostgreSQL full-text	grow. to start. Upgrade when listings cheaper Good enough for MVP and	unnecessary early cost. Advanced search may add
Payments	payment provider Stripe or local trusted	receipts, transaction records. supports payment flows, Reduces security risk and	handling from scratch. Do not build custom payment
AI	No core AI model for MVP	improvement. detection, listing text category suggestions, scam Add AI later for moderation, transactions matter more first. Trust, supply, demand, and	modern. Do not add AI just to look

secure enough for real users, simple to operate, and scalable later.
Stack principle: choose technology because it supports the business goal: fast to build,